

CWM Foundations of Digital Infrastructure

Exercise 6: Mini Project

Introduction

In this exercise, you will develop a mini project of your choice. The project does not need to be complex, but should demonstrate the concepts taught during the CWM.

You are required to submit a pdf file describing the project, its design and implementation, and any performance tests (if relevant). You also need to submit any written code and generated log files, which should be linked to your repository. See below instructions for the use of AI as part of the project.

The project must demonstrate the concepts taught as part of this CWM, and the report needs to reflect on those in the context of the project.

Project ideas

1. Fuzz Tester

Fuzz testing is an automated software testing technique that injects invalid, malformed, or random data into a program to find security vulnerabilities and bugs. Implement a fuzz testing programme that will test the code you developed during the CWM (and potentially other programmes too). Report any bugs or vulnerabilities found.

2. Performance Bottleneck Detector

Implement an automated tool that collects performance metrics during an application run time (using tools used in previous exercises or others), and detects and reports performance issues. As a second stage, the tool can propose solutions to the detected problems.

3. Disaster Resilience Index / Digital Equity Index

Use public datasets and/or measurements to evaluate the quality of digital infrastructure in different places in the UK / worldwide.

Use the information to define one of the following:

- Disaster resilience index – how likely is the infrastructure to fail in case of a disaster, such as a storm or a cyber-attack.
- Digital equity index – what is the quality of the digital infrastructure available to residence in the area.

Compare the index across different regions, and propose mitigations and ways to improve the local digital infrastructure.

4. Network sustainability tracker

Implement a solution that tracks the volume of network traffic generated by an application, and uses it to account for the energy consumption of the program. Based on the results of exercise 3, assume a linear power model, and convert the traffic through the network interface (per second/minute) into energy consumption. An advanced version will track multiple applications (traffic flows).

5. Machine learning sustainability estimator

Implement a solution that given learns the energy efficiency of different ML models, and given a task uses the most energy efficient ML model suitable for the task. At the first stage, only report the ML performance, energy consumption and carbon

emissions for all models. At the second stage, select the model that will best balance ML performance and emissions, considering also carbon intensity.

These are just a few ideas; you can come up with your own project idea!

Please consult with the CWM's lead Professor before starting your project, regardless of if it is from the proposed project ideas or your own idea.

Use of AI

The use of AI is permitted as part of this assignment, and must comply with the following guidelines:

- AI must NOT be used to write the entire project, or any parts of the report.
- Students are evaluated based on the quality and scope of the work they have done themselves (not by AI), and the quality of their critical thinking and discussion.
- The use of AI (or no use of AI) must be declared. Marks will be reduced if such a declaration is missing, even if AI was not use.
- The AI system used must be declared, including the provider, model, and version where available
- The report must indicate the specific purposes for which AI was used and identify the components of the project that were assisted by AI.

- You must include in your repository a complete prompt and response log for all AI interactions that materially influenced the project.
- AI should not be used to generate graphs directly. AI can be used to write functions that will generate graphs.
- AI should not be used for data analysis. AI can be used to write functions that will be used for data analysis.
- AI should not be used for any discussion parts.
- You must be able to explain any code written by AI.
- Any discovered hallucinations (e.g., citations, data) will result in a significant loss of marks.
- It is allowed to use AI to improve the phrasing and quality of writing of the report.
- Sources of data used by the AI must be referenced in the report (where the AI reports such sources). Similarly, all datasets must be referenced and links included.
- Further guidelines and restrictions for the use of AI may be provided by the course's team.
- The use of AI must further comply with the University's use of AI policy and departmental guidelines.

Additional submission instructions are at the beginning of this document.